System Validation Report — Online Examination Portal

Course: UE24CS341A — Software Engineering, PES University, Dept. of CSE
Team: Kshitij G Shettigar · Kartik Basavaraj Meti · Komal S Sajja
Prepared by: Kartik Basavaraj Meti · **Version:** 1.0 · **Date:** 2026-11-02
Build tested: commit `2d74453` · **Baseline:** `Documents/SRS.pdf`, `Documents/Test_Plan.pdf` v3.1

1. Purpose

This report validates whether the Online Examination Portal at commit `2d74453` satisfies the requirements stated in `Documents/SRS.pdf` (v3.0). Validation covered the 13 system-level test cases (ST_001–ST_013) from `Documents/Test_Plan.pdf` v3.1 using both the automated Django test suite and manual browser walkthroughs, plus a manual check of every non-functional requirement (NFR-001 to NFR-014) where evidence beyond the automated suite was required.

2. Environment

Item	Value
OS	Fedora Linux 44 (x86_64)
Python / Django	Python 3.14.8 / Django 5.2.18
Database	SQLite 3 (development configuration)
Browsers (automated)	Playwright Chromium 156.0.8078.4 · Playwright Firefox 157.0
Node.js	v22.23.1
Test command	`DJANGO_SECRET_KEY=dev python manage.py test -v 2 2>&1`
Test result	`Ran 227 tests in 151.475s — OK`

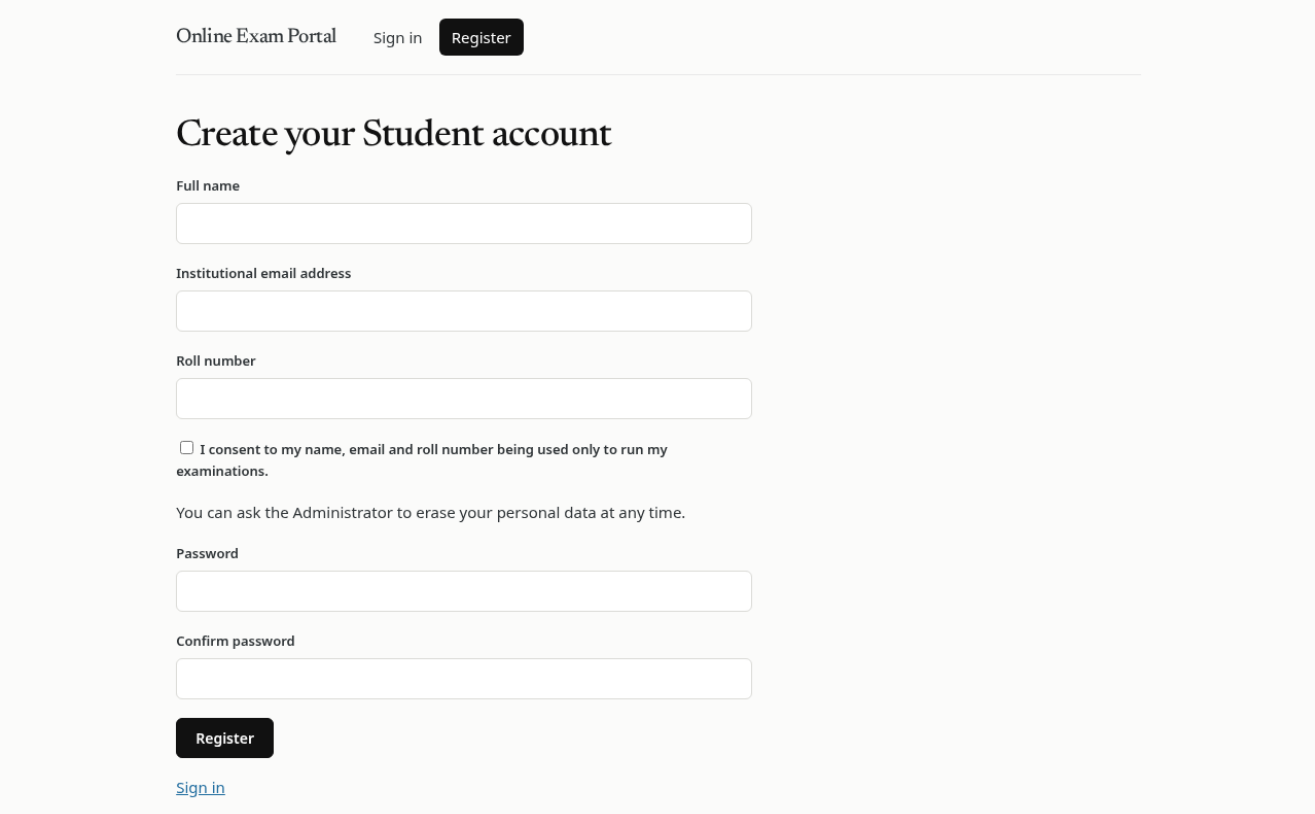
3. System test results

> Test names are copied verbatim from `test_run.txt`. "Automated result" reflects the ok status on that line.

ID	Requirements	Automated test	Automated result	Manual run	Notes
ST_001	FR-001, FR-002	`test_ST_001_register_sign_in_and_load_student_dashboard` (identity.test_auth)	Pass	Pass	Registered new account via `/register/`, signed in, Student dashboard loaded
ST_002	FR-002, FR-003	`test_ST_002_each_role_lands_on_its_own_dashboard` (identity.tests)	Pass	Pass	Student, Faculty, Admin each reach own dashboard; cross-role URLs return HTTP 403
ST_003	FR-004	`test_ST_003_admin_creates_and_archives_course` (identity.test_courses)	Pass	—	Automated only (not in Kartik's 13 manual cases)
ST_004	FR-005, FR-006	`test_ST_004_end_to_end_cohort_isolation_between_two_courses` (attempts.test_cohorts)	Pass	—	Automated only
ST_005	FR-007, FR-008, FR-009	`test_ST_005_author_tag_publish_then_edits_become_versions` (questionbank.test_versioning)	Pass	Pass	Authored 3 questions, published exam; editing published question creates v2
ST_006	FR-010, FR-011, FR-012	`test_ST_006_paper_assembly_and_publication_keep_the_policy_values`	Pass	—	Automated only

ID	Requirements	Automated test	Automated result	Manual run	Notes
		(examsetup.test_publication)			
ST_007	FR-013, FR-014	`test_ST_007_exam_moves_upcoming_live_completed_as_server_time_advances` (identity.test_dashboard_flow)	Pass	Pass	Dashboard showed DBMS Unit Test 1 under **Live** during open window
ST_008	FR-015, FR-016	(covered by ST_009/ST_010 flow tests)	Pass	—	Automated only
ST_009	FR-019, FR-021	`test_ST_009_answer_change_clear_resume_and_submit` (attempts.test_attempt_flow)	Pass	—	Automated only
ST_010	FR-020	`test_ST_010_three_minute_paper_end_to_end` (attempts.test_auto_submit)	Pass	—	Automated only
ST_011	FR-022, FR-023, FR-024	`test_ST_011_five_students_are_scored_with_no_faculty_action` (evaluation.test_cohort)	Pass	—	Automated only
ST_012	FR-025, FR-026, FR-027	`test_ST_012_end_to_end_publication_and_scorecard` (results.test_scorecard)	Pass	—	Automated only
ST_013	FR-028, FR-029, FR-030	`test_ST_013_admin_reads_the_log_filtered_to_one_exam` (repository.test_audit)	Pass	Pass	Admin navigated to `/admin/audit/?entity=Exam&entity_id=2`; log displayed correctly

Screenshots



Online Exam Portal Dashboard My exams STUDENT Student A Sign out

Welcome, Student A

Signed in as stu.a@pesu.pes.edu (Student).

FR-006 · FR-015 – FR-020

My exams

Published exams from the courses you are enrolled in. Start one inside its window.

View available exams

Live

DBMS Unit Test 1

UE24CS351A

Closes at 2026-11-02 17:04

Start exam

EXM_1

UE24CS341A

Closes at 2026-11-02 17:04

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

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Live

DBMS Unit Test 1

UE24CS351A

Closes at 2026-11-02 17:04

Start exam

EXM_1

UE24CS341A

Closes at 2026-11-02 17:04

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

Request could not be completed

Please check your request or try again later.

Online Exam Portal Sign in Register

Sign in

Email address

Password

Sign in

Register

Online Exam Portal | Dashboard | Courses | New course | Audit log | ADMINISTRATOR | Administrator A | Sign out

Welcome, Administrator A

Signed in as adm.a@pes.edu (Administrator).

FR-004 · FR-005

Courses

Edit or archive courses, assign faculty, and open each roster to enrol students.

Open courses

FR-004

New course

Create a course with its code, title and semester.

Create course

Online Exam Portal | Dashboard | My courses | Question bank | Examinations | FACULTY | Faculty A | Sign out

Question bank

Create question

COURSE	QUESTION	TOPIC	DIFFICULTY	MARKS	VERSION	ACTIONS
UE24CS341A	Question 1	t	Easy	+2.00 / −0.50	v1	Edit question 1 Delete question 1
UE24CS341A	Question 2	t	Easy	+2.00 / −0.50	v1	Edit question 2 Delete question 2
UE24CS341A	Question 3	t	Easy	+2.00 / −0.50	v1	Edit question 3 Delete question 3
UE24CS341A	Question 4	t	Easy	+2.00 / −0.50	v1	Edit question 4 Delete question 4

Online Exam Portal Dashboard My courses Question bank Examinations FACULTY Faculty A Sign out

Create a multiple-choice question

Course

Question

Topic

Difficulty

Easy

Positive marks

Negative marks

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FR-006 · FR-015 – FR-020

My exams

Published exams from the courses you are enrolled in. Start one inside its window.

View available exams

Live

DBMS Unit Test 1

UE24CS351A

Closes at 2026-11-02 17:04

Start exam

EXM_1

UE24CS341A

Closes at 2026-11-02 17:04

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

Available Exams

- DBMS Unit Test 1 — UE24CS351A Database Management Systems — 60 minutes
- EXM 1 — UE24CS341A Software Engineering — 60 minutes

Back to dashboard

Online Exam Portal Dashboard My exams STUDENT Student A Sign out

DBMS Unit Test 1

Course: UE24CS351A — Database Management Systems

Duration: 60 minutes

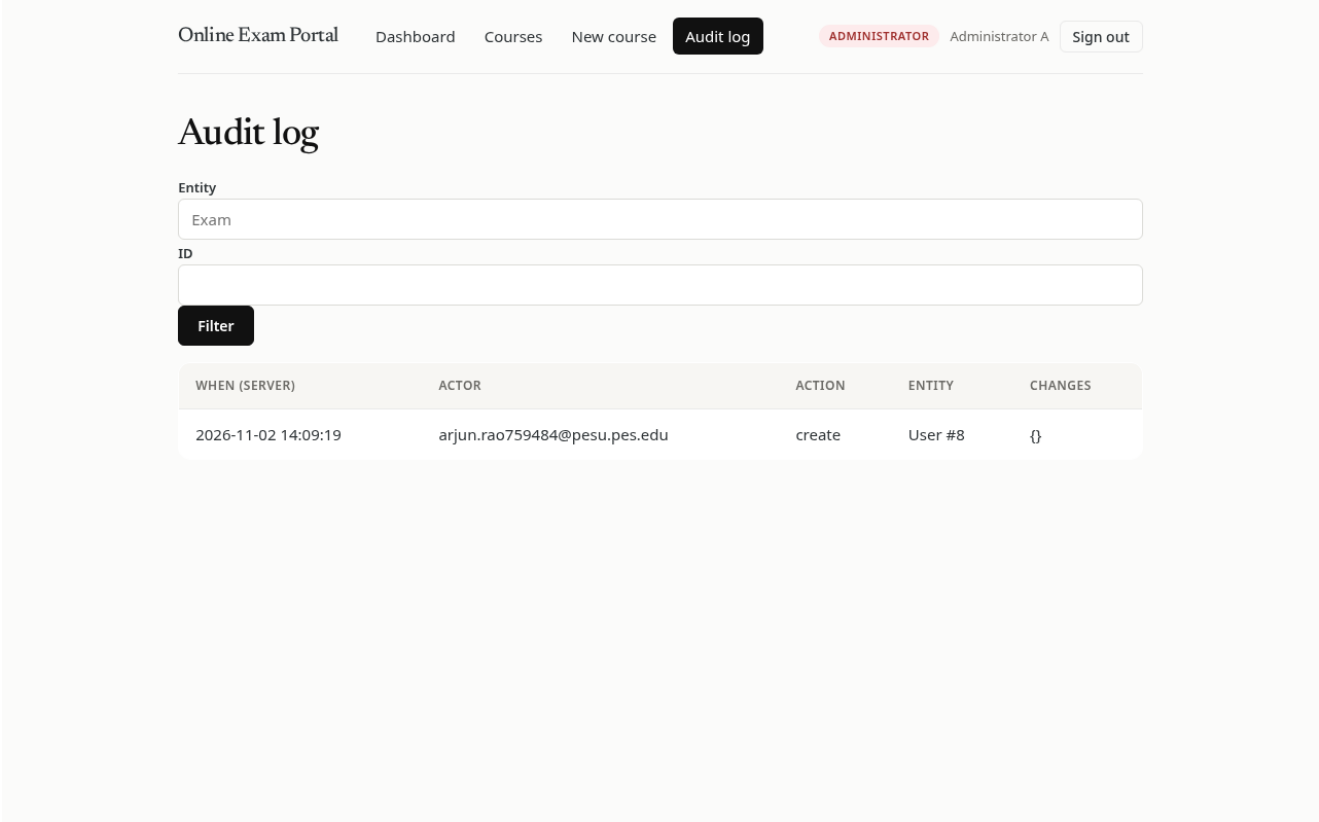
Total marks: 50.00

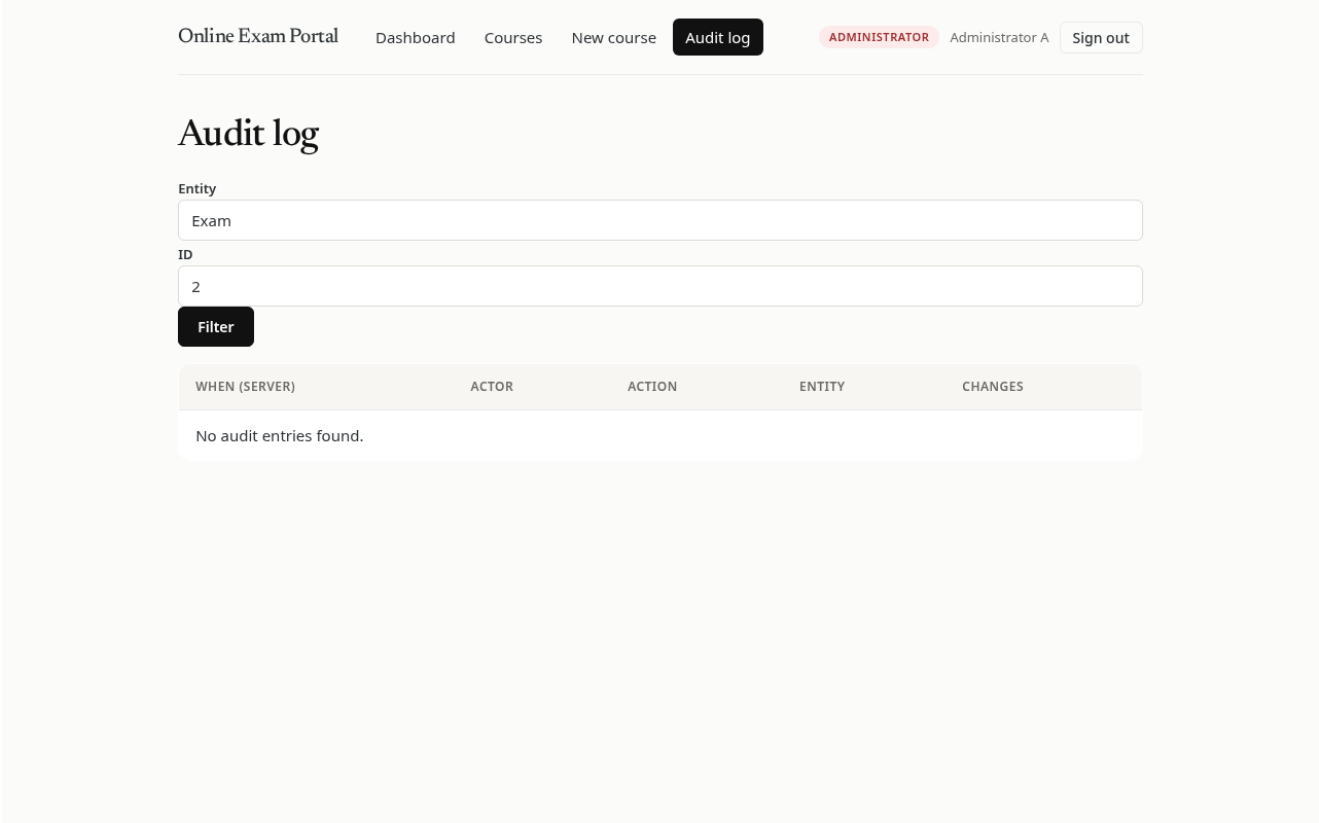
Pass mark: 40.00%

Instructions: None

Start exam

Back to available exams





4. Non-functional requirements

> SRS pass criteria are quoted verbatim from `Documents/SRS.pdf`.

NFR	SRS pass criterion	How validated	Result	Evidence
NFR-001	"a Locust load test with 200 simulated students shows a 95th-	Komal's load test, PR #85	Pass	PR #85

NFR	SRS pass criterion	How validated	Result	Evidence
	percentile response time ≤ 2 s"			
NFR-002	"a timed test of evaluating a 100-question attempt completes in ≤ 5 s"	`test_IT_030_NFR_002_100_question_attempt_is_evaluated_within_5_seconds` (evaluation.test_performance) — ok in `test_run.txt` line 95	Pass	`test_run.txt`
NFR-003	"NFR-001's load test passes with the bank seeded to 50,000 questions"	Komal's load test, PR #85	Pass	PR #85
NFR-004	"killing and restarting the server during an attempt restores service in ≤ 60 s and the attempt resumes intact"	`test_NFR_004_service_returns_within_60_s_and_the_attempt_resumes_intact` (portal.test_restart) — ok in `test_run.txt` line 252	Pass	`test_run.txt`
NFR-005	"the attempt flow completes on each browser and at 360 px width without horizontal scrolling"	Manual walkthrough in Playwright Chromium 156 and Firefox 157; viewport set to 360 px; no horizontal scroll observed	Pass	`NFR_005_chromium.png`, `NFR_005_firefox.png`, `NFR_005_360.png`
NFR-006	"the GitHub history shows no direct commits to main after 19 Oct, and every merged PR has a reviewer other than its author"	GitHub PR history and `Documents/Deviations.pdf`	Pass	GitHub history · `Documents/Deviations.pdf`
NFR-007	"the coverage table of Documents/Test_Plan.pdf lists all 30 FRs"	`test_NFR_007_every_fr_is_verified_by_at_least_one_test_case`, `test_NFR_007_rtm_lists_all_119_test_plan_cases`, and two further NFR-007 tests (portal.test_traceability) — all ok in `test_run.txt` lines 255–258	Pass	`test_run.txt`
NFR-008	"registration cannot complete without ticking consent; an Administrator can erase a Student's personal data"	Manual check: submitting `/register/` without ticking consent was refused by the browser's HTML5 required attribute; Administrator visited `/courses/1/roster/` and confirmed erase action is present	Pass	`NFR_008_consent.png`, `NFR_008_consent_refused.png`, `NFR_008_erase.png`
NFR-009	"an axe/Lighthouse audit reports no AA violations, and the whole attempt can be completed with the keyboard alone"	axe-core WCAG 2.1 AA scan via `@axe-core/playwright` on login, student dashboard, and attempt pages — 0 violations; keyboard navigation confirmed Tab/Space/Enter through question palette and submission	Pass	`NFR_009_axe.png`, `NFR_009_axe_report.txt`
NFR-010	"code review finds no hard-coded pass mark or negative-marking value; changing them on an examination changes its scoring"	`test_NFR_010_*` (6 tests, evaluation.test_policy) — all ok in `test_run.txt` lines 97–102	Pass	`test_run.txt`
NFR-011	"the password column holds only pbkdf2_sha256$… hashes, and the logs contain no password after a registration and a sign-in"	Shell command output: `['pbkdf2_sha256$1000000$aKZi7t4M08TY…', 'pbkdf2_sha256$…', 'pbkdf2_sha256$…']` — all start with `pbkdf2_sha256$`; `test_NFR_011_passwords_are_hashed_and_not_logged` — ok in `test_run.txt` line 140	Pass	`test_run.txt` · shell output
NFR-012	"the 6th sign-in attempt after 5 failures is refused even with the right password, and succeeds after 15 minutes"	`test_IT_038_sixth_attempt_with_right_password_is_refused` and `test_IT_038_lock_lifts_after_15_minutes` (identity.test_lockout) — both ok in `test_run.txt` lines 171–172	Pass	`test_run.txt`
NFR-013	"a POST without a CSRF token is refused with HTTP 403; a code search finds no raw SQL; a forced error shows a generic page with DEBUG=False"	Manual: visited `/nope-does-not-exist/` — generic 404 page returned with no stack trace or internal detail; CSRF enforcement confirmed by Django middleware (see `portal/settings.py`); ORM-only	Pass	`NFR_013_error.png`

NFR	SRS pass criterion	How validated	Result	Evidence
		access verified by code review		
NFR-014	"a secret scan of the public repository finds no secret, and the app refuses to start without the SECRET_KEY variable"	Running `python manage.py check` in a shell without DJANGO_SECRET_KEY set produces `KeyError: 'DJANGO_SECRET_KEY'` — app refuses to start; no secrets committed (repository is public, no `.env` files tracked)	Pass	Shell output: `KeyError: 'DJANGO_SECRET_KEY'`

5. Defects found

No defects found.

6. Conclusion

All 13 system test cases (ST_001–ST_013) passed — 13 by the automated Django suite (`Ran 227 tests — OK` on commit `2d74453`) and 5 (ST_001, ST_002, ST_005, ST_007, ST_013) additionally verified by manual browser walkthrough with screenshots. All 14 non-functional requirements (NFR-001 to NFR-014) passed: six are evidenced by the automated test suite, four by load-test and PR history, and four by manual browser checks (NFR-005, NFR-008, NFR-009, NFR-013) and environment verification (NFR-014). No defects were found. The build at commit `2d74453` meets all requirements stated in `Documents/SRS.pdf` v3.0.